

Nucleic Acids

Current 26-page faculty deck primary - official Week 4 objectives integrated - organized for high-yield retrieval

5 official objectives

DNA melting

Gyrase

Telomeres & repeats

Chromatin remodeling

RNA classes

SOURCE PRIORITY: the uploaded current faculty deck controls terminology, examples, and emphasis. The official Week 4 learning-objective sheet defines what must be mastered. This companion reorganizes the same material for retrieval and application without replacing the source deck.

Official Week 4 Objectives

1. Explain nucleic-acid denaturation and renaturation (annealing), including how these processes are used in laboratory hybridization techniques.
2. Discuss salient features of prokaryotic DNA and the importance of DNA gyrase.
3. Discuss salient features of eukaryotic DNA and distinguish telomeres, repetitive versus non-repetitive sequences, heterochromatin, and euchromatin.
4. Describe molecular mechanisms that switch DNA between heterochromatin and euchromatin and summarize chromatin remodeling required for DNA accessibility.
5. Discuss structure and function of RNAs common to prokaryotes and eukaryotes and the additional functional/regulatory RNAs found in eukaryotes.

Start with the invariant rules

- DNA and RNA are heteropolymers of nucleotides joined by phosphodiester bonds.
- Nucleic-acid sequences are written 5' to 3'.
- Complementary strands pair only when they are antiparallel.
- Denaturation separates paired strands and disrupts helical structure without breaking the covalent phosphodiester backbone.

Denaturation

Favored by increased temperature, extreme/high pH, low ionic strength, and chemical denaturants. Duplex disruption produces a **hyperchromic shift**: increased absorbance at 260 nm.

Renaturation / annealing

Complementary strands reassociate when conditions again favor base pairing. Gradual cooling and higher salt favor annealing because ionic shielding stabilizes the duplex.

T_m AND GC CONTENT

T_m is the temperature at which 50% of the helix is denatured. Higher GC content increases T_m.

STRINGENCY

High stringency = high temperature + low salt and therefore requires close complementarity.
Low stringency = lower temperature + higher salt and allows related but imperfectly matched sequences to hybridize.

Why hybridization matters: a labeled nucleic-acid probe can identify complementary or related sequences. Because related sequences can still anneal under lower-stringency conditions, hybridization is generally a poor stand-alone method for distinguishing a single-base mutation or a very small insertion/deletion.

Slide 6

Prokaryotic DNA and gyrase

LO 2

Architecture

Bacterial chromosomal DNA is covalently closed, circular, relatively "naked," and maintained in a **negatively supercoiled** state.

Gyrase

DNA gyrase is a bacterial **type II topoisomerase**. It makes a transient double-strand break, passes DNA through the break, reseals it, and uses ATP.

Gyrase creates negative supercoils

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DNA packing improves

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Replication/transcription can proceed

Drug connection: ciprofloxacin and novobiocin target gyrase. The lecture emphasizes that these drugs interfere after DNA cleavage and before resealing, so DNA double-strand breaks accumulate.

Slides 7-9

The eukaryotic genome: linear DNA, telomeres, and repetitive sequences

LO 3

- Nuclear chromosomes are linear; mitochondrial DNA is circular.
- Most eukaryotic genes are segmented into **exons** retained in mature mRNA and **introns** removed during splicing.
- Chromosome ends contain a 3' overhang and long stretches of repeated hexameric telomere sequence.

Sequence class

Pattern

High-yield significance

Non-repetitive	Single/unique sequence per haploid genome	Major protein-coding fraction discussed in lecture
Highly repetitive	Short sequences repeated many times in tandem	Centromeres/telomeres; generally transcriptionally inactive
Moderately repetitive	Dispersed copies including LINEs and SINEs	Mobile-element/retrotransposon relationships and genome evolution
Minisatellites	Usually 9-30 bp units repeated 10-60 times	VNTR polymorphism; historically useful in identity/paternity analysis
Microsatellites	2-6 bp units repeated in tandem	STR polymorphism; trinucleotide repeats can expand and cause disease

Repeat-size anchor: LINE = long interspersed element; SINE = short interspersed element; VNTR = minisatellite-style variable number of tandem repeats; STR = microsatellite-style short tandem repeat.

Slides 10-14

Chromatin is DNA plus histones, and it is dynamic

LO 3-4

Histones are basic, lysine/arginine-rich proteins. Two copies each of H2A, H2B, H3, and H4 form the histone octamer; roughly 150 bp of DNA wraps around the core to form a **nucleosome**. Linker DNA between nucleosomes associates with H1. Scaffold proteins support higher-order compaction.

Euchromatin

Open/lightly condensed, accessible, and generally transcriptionally active.

Heterochromatin

Highly condensed, less accessible, and generally transcriptionally silent.

Key concept: the euchromatin/heterochromatin state is reversible and can respond to hormones, nutrients, drugs, xenobiotics, metals, and infectious stimuli. These changes can be inherited through somatic cell divisions without changing the DNA sequence itself.

Modification	Enzyme	Typical chromatin effect	Why
CpG cytosine methylation	DNA methyltransferases (DNMTs)	Compaction and gene silencing	Methylated promoter CpG islands become less accessible to transcription machinery
Histone lysine acetylation	Histone acetyltransferases (HATs)	Open chromatin and gene activation	Acetylation removes lysine's positive charge and weakens histone-DNA attraction
Histone deacetylation	Histone deacetylases (HDACs)	Condensation and repression	Restores stronger electrostatic interaction with DNA
Histone methylation	Histone methyltransferase pathways	Context dependent	Lecture examples: H3K4 trimethylation is activating; H3K27 trimethylation is repressive

Clinical anchors from the deck: decitabine is presented as a DNMT inhibitor; valproic acid is presented as an HDAC inhibitor. In general, inhibiting DNMTs or HDACs tends to favor expression, whereas inhibiting HAT activity tends to favor repression.

Methylation closes; acetylation opens. Do not apply that shortcut to histone methylation, which is residue-specific.

RNA is usually single stranded, contains ribose and U instead of T, and folds through intramolecular base pairing. RNA can carry information, form structural scaffolds, regulate gene expression, and catalyze reactions.

RNA	Lecture anchor	High-yield features
tRNA		

	Adaptor for translation	75-85 nt; many modified bases; cloverleaf secondary structure; L-shaped tertiary structure; 3' CCA amino-acid attachment site; anticodon at opposite end
rRNA	Structural/catalytic core of ribosome	Most abundant RNA; prokaryotes 23S/16S/5S; eukaryotes 28S/18S/5.8S/5S; 23S/28S peptidyl-transferase function
mRNA	Protein-coding information	Least abundant major RNA; short half-life; eukaryotic mRNA has 5' 7-methylguanosine cap and 3' poly(A) tail; bacteria are commonly polycistronic, eukaryotic mRNAs generally monocistronic

Functional eukaryotic RNAs

hnRNA = mRNA precursor. **snRNA** = nuclear, U-rich RNA used in pre-mRNA maturation/splicing. **scRNA** = cytosolic RNA associated with proteins involved in signal-peptide recognition.

Regulatory eukaryotic RNAs

lncRNA >200 nt; can recruit chromatin-remodeling machinery. Xist silences the X chromosome from which it is transcribed. **miRNA** ~19-25 nt and mediates post-transcriptional silencing by decreasing mRNA stability and/or translation.

RNA is not just a messenger. The lecture explicitly uses RNA as information (mRNA), adaptor (tRNA), structure/catalyst (rRNA), processing machinery (snRNA), and regulator (miRNA/lncRNA).

Master map

One-pass retrieval map

All LOs

If the question gives...	Think...
High temperature / low salt	High stringency; only highly complementary strands remain paired
Bacterial circular DNA + antibiotic	Gyrase / type II topoisomerase
Promoter CpG methylation	Heterochromatin tendency and transcriptional silencing
Histone acetylation	Weaker DNA-histone attraction and open chromatin
X-chromosome silencing	Xist lncRNA
	miRNA

Post-transcriptional silencing by a 19-25 nt RNA	
Peptide bond formation in the ribosome	rRNA ribozyme activity

Exam traps

Details worth protecting from mix-ups

All LOs

Trap 1: Denaturation breaks hydrogen-bonding/base-stacking interactions, not the phosphodiester backbone.

Trap 2: High stringency is not "high salt." It is higher temperature and lower salt.

Trap 3: DNA methylation generally represses promoter activity, but histone methylation is not uniformly activating or repressing.

Trap 4: Acetylation of lysine removes its positive charge; lysine methylation does not.

Trap 5: Eukaryotic mRNA cap and poly(A) tail are post-transcriptional modifications.

30-SECOND FINISH

Gyrase = bacterial type II topo. Telomeres/repeats distinguish key eukaryotic chromosome features. Euchromatin is open; heterochromatin is closed. DNMT closes; HAT opens; HDAC closes. tRNA adapts, rRNA builds/catalyzes, mRNA encodes, Xist silences, miRNA suppresses after transcription.